Charles Clement Lehnen

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Self-sufficient quantitative ecologist and conservation biologist passionate about rewilding, translating ecological principles into management decisions, and the cascading ecological effects of species interactions. Experienced in independently designing, funding, and executing large-scale, internationally collaborative field projects from conception to publication. Skilled in advanced statistical and geospatial analyses, scientific communication, and mentorship.

Education

August 2021 - May 2027 — **Doctoral Candidate**. *University of Southern California (USC); Department of Biological Sciences (BISC)*
Advisor: Dr. Craig Stanford
Provost Fellow; Wrigley Institute Graduate Fellow; Gold Family Fellow; JEP STEM Education Fellow; Graduate Certificate in Spatial Analytics

September 2008 - May 2012 — **Bachelor of Science.** *University of Minnesota (UMN); College of Biological Sciences (CBS)*
Biology; Minors: Mathematics, Spanish
Honors Program; CBS Dean's Scholar Leadership Program

Selected Employment

August 2021 - May 2027 — **Doctoral Candidate**. *University of Southern California (USC); Department of Biological Sciences (BISC)*
Advisor: Dr. Craig Stanford

February 2020 - August 2021 — **Subterranean Technician.** *Bat Conservation International (BCI); U.S. Bureau of Land Management (BLM); New Mexico Abandoned Mine Land (AML) Program*
Supervisors: Shawn Thomas; Jackson Bain

June 2016 - March 2020 — **Researcher I; Visiting Scientist.** *University of Minnesota (UMN); Charles Darwin Foundation (CDF)*
Advisors: Dr. Charlotte Causton; Dr. George Heimpel; Dr. Bradley Sinclair

March - May 2017 — **Research Associate.** *Great Basin Institute (GBI); U.S. Fish and Wildlife Service (USFWS); AmeriCorps*
Supervisors: Linda Allison; Terry Christopher

June - August 2016 — **GIS GeoPortal Technician.** *Charles Darwin Foundation*
Supervisors: Byron Delgado; Jesus Jimenez

December 2015 - May 2016 — **Galapagos National Park Volunteer.** *Galapagos National Park; Centro de Crianza de Tortugas Terrestres Jacinto Gordillo*
Supervisors: Victor Diaz; Franklin Chicaiza; Jeffreys Málaga

References upon request *October 2026*

September 2012 - September 2015	**Assistant Scientist; Researcher II.** *Masonic Cancer Research Center, UMN Medical School Department of Pediatrics, Division of Bone Marrow Transplantation* Advisors: Dr. Bruce Blazar; Dr. Keli Hippen
August 2011 - August 2012	**Undergraduate Research Assistant, Summer Field Manager.** *UMN Department of Horticultural Sciences; Urban Forestry Outreach, Research, and Extension (UFore) (FKA Teaching, Research and Extension (TRE) Nursery)* Advisors: Chad P. Giblin, M.S; Dr. Jeff Gillman
September 2009 - August 2010	**Undergraduate Laboratory Assistant.** *Masonic Cancer Research Center, UMN Medical School Department of Pediatrics, Division of Bone Marrow Transplantation* Advisors: Dr. Bruce Blazar; Dr. Keli Hippen

Teaching Experience

Fall 2026	**Teaching assistant**. *USC* Advanced General Biology: Organismal Biology & Evolution *(BISC 121) - undergrad* Supervisors: Dr. David Hutchins
Spring 2023, 2024, 2025, & 2026	**Teaching assistant and course developer**. *USC* Ecology and Biodiversity *(BISC 499; BISC 404) - 46 undergrad & grad* Supervisors: Dr. Laura Melissa Guzman; Dr. Cameron Egan; Dr. Trond Sigurdsen *BISC Outstanding TA Award (2023)* ***USC Outstanding TA Award (University wide; 2026)***
Spring 2026	**Field teaching assistant.** *Middlebury College* The Living Deserts of North America **(***BIOL 0445) - 10 undergrad* Supervisors: Dr. Kirsten Coe *Publication "Living Deserts: Blueprint for a field-based experiential learning undergraduate course" in prep*
Fall 2023	**Instructor.** *Joint Educational Project (JEP); USC* WonderKids - *40 K-5* Supervisors: Dr. Dieuwertje Kast; Jessica Stellman *Publication: "Zooming into STEM: Designing Curriculum Around Guest Presentations in an Online After-School STEM Program for Title I Elementary Students" Innovations in Science Teacher Education. 2026*
Fall 2021 & 2022 Spring 2022	**Teaching assistant**. *USC* The Human Animal *(HBIO 200) - 118 undergrad*

Supervisors: Dr. Craig Stanford; Dr. Maureen McCarthy

BISC Outstanding TA Award

August - December 2011 **Teaching assistant.** *UMN*
Human Physiology *(PHSL 3051) - undergrad*

Submitted or in preparation publications

***Lehnen, Charles**, Bradley J. Sinclair, Paola F. Lahuatte, Rebecca A. Boulton, Ma. Beatriz Pérez-Lara, Alejandro Mieles, Charlotte E. Causton, George E. Heimpel. 2026.* "Introduced flies outnumber native and endemic flies in larval resources across islands, seasons and vegetation zones in the Galapagos Archipelago." *In preparation: 98% (36 pp.) / planned journal: Diversity and Distributions / submission in October 2026.*

*Coe, Kirsten K.**, Charles Lehnen**. "Living Deserts: Blueprint for a field-based experiential learning undergraduate course" In preparation: 90% (25 pp.) / planned journal: Journal of College Science Teaching / submission by November 2026.*

*Yan Yin Cheung, Vaughn M. Shirey, Teagan M. Baiotto, Brian V. Brown, Austin J. Baker, Jayme Lewthwaite, **Charles Lehnen**, Terrence P. McGlynn, Emily Hartop, Peter H. Kerr, Eric Wood, and Laura Melissa Guzman**.* "The built environment shapes synchrony in an insect community more than climate." *In preparation: 80% (27 pp.) / submission in Fall 2026.*

*Menzel, Frank, Adam Broadley, Werner Mohrig, **Charles Lehnen**, Heinke Jäger & Ellen Kauschke. "*Black fungus gnats (Diptera: Sciaridae) of the Galápagos Islands (Ecuador)" *In preparation: 90% (45 pp.) / planned journal: Zootaxa / submission by December 2026.*

***Lehnen, Charles,** Craig Stanford.* "Detecting Early Vegetation Responses to Tortoise Rewilding across Santa Fe Island" *In preparation: 80% (28 pp.) / planned journal: TBD / submission Spring 2027*

***Lehnen, Charles,** Craig Stanford.* "Activity Patterns of Giant Galápagos Tortoises Over Time and Space" *In preparation: 80% (22 pp.) / planned journal: TBD / submission Spring 2027*

Peer Reviewed Publications

***Lehnen, Charles**, Jessica Stellmann, Andrea Morfin Valencia, Dr. Dieuwertje Kast. 2026.* "Zooming into STEM: Designing Curriculum Around Guest Presentations in an Online After-School STEM Program for Title I Elementary Students" *Innovations in Science Teacher Education* *https://doi.org/10.5281/zenodo.21448934*

*Lewthwaite, Jayme; Teagan Baiotto; Brian Brown; Yan Yin Cheung; Austin Baker, **Charles Lehnen**, Terrence McGlynn. 2024.* "Drivers of arthropod biodiversity in an urban ecosystem." *Scientific Reports 14, no. 1: 390.*

*Hippen, Keli L., Benjamin Watkins, Victor Tkachev, Amanda M. Lemire, **Charles Lehnen**, Megan J. Riddle, Karnail Singh et al. 2016* "Preclinical Testing of Antihuman CD28 Fab′ Antibody in a Novel Nonhuman Primate Small Animal Rodent Model of Xenogenic Graft-Versus-Host Disease." *Transplantation, 100(12), p.2630.*

Other publications

*Guzman, Melissa; **Charles Lehnen**; Teagan Baiotto. 2024 "Bugs thrive in urban Los Angeles – volunteers' traps reveal biodiversity hot spots for city insects and spiders" The Conversation (Republished by 13 outlets) DOI 10.64628/AAI.yeq944dqv*

***Lehnen, Charles**, David J Matthes. 2012. "KIAA0922" Wikipedia*

Grants & Fellowships

April 2026	USC Graduate School Summer Research and Writing Grant	$2,000
Spring 2022 - present	USC Jane Goodall Fund	$7,165 total
May 2024	Lewis and Clark Fund for Exploration and Field Research Scholarship	$4,900
May 2024, May 2023, May 2022	USC Dornsife PhD Academy Scholarship	$1,500 total

June 2024	**Fulbright Fellowship alternate**	
April 2024	**Gold Family Fellowship**	$5,000
April 2024	USC Graduate Student Government (GSG) Professional Development Fund	$500
March 2024	USC Graduate School Travel/Research Award	$1,000
January 2024	Iguanas in the Balance Grant	$8,500
August - December 2023	**USC Joint Educational Project (JEP) STEM Fellowship**	$17,850
April 2023	Chelonian Research Foundation (CRF) Congdon & Dickson Research Fund (CDRF)	$3,000
January 2023	**USC Wrigley Institute Graduate Fellowship**	$6,000
October 2022	USC External Fellowship Bootcamp Award	$1,000
August 2021 - August 2026	**USC Provost Fellowship**	$17,500 total
April 2021	The Saint Paul Festival & Heritage Foundation Ambassadors Scholarship	$500

Honors & Awards

July 2026	North American Congress for Conservation and Restoration (NACCR) "Student Presentation Award" finalist	
March 2026	Second place at 14th annual USC Graduate Student Government (GSG) Graduate Research Symposium	$100
April 2026	**USC Outstanding TA Award (University wide)**	$1,000
April 2025	USC Dornsife College of Letters, Arts, and Sciences Ph.D. Academy Certificate	
April 2023	USC Biological Sciences (BISC) Outstanding TA Award	$200
Spring 2013	UMN Tony Diggs Excellence Award - Outstanding Student & Community Relations	
2008 - 2012	UMN Honors Program	

2008 - 2012	UMN College of Biological Sciences (CBS) “Dean’s Scholars” Leadership in Science Program
Spring 2008	The Degree of Honor Foundation Salute to Youth award
2007 - 2008	Minnesota Postsecondary Enrollment Options (PSEO) scholar
January 2008	Boy Scouts of America Eagle Scout Award
Spring 2007	Minnesota High School League (MSHSL) ExCEL award
Fall 2006	PSAT Commended Student

Invited Presentations

Science & Suds

2026 ***Charles Lehnen****, Dr. Craig Stanford, Dr. Washington Tapia.* Slow and Steady Saves the Island: When Giant Galápagos Tortoises Return. Los Angeles, California. *Oral presentation*

Loma Linda University Department of Earth and Biological Sciences Biology Seminar.

2026 ***Charles Lehnen****, Dr. Craig Stanford, Dr. Washington Tapia.* Assessing Española Giant Tortoise introduction impact on Santa Fe Island with remote sensing and behavioral models. Linda Loma, California. *Oral presentation*

Turtle Conservancy

2026 ***Charles Lehnen****, Dr. Craig Stanford, Dr. Washington Tapia.* Assessing Española Giant Tortoise introduction impact on Santa Fe Island with remote sensing and behavioral models.. Ojai, California. *Oral presentation.*

USC PhD Academy: The Ph.D. Elevator Pitch: Communicating Your Scholarship with Confidence

2025 ***Charles Lehnen*** Los Angeles, CA, USA. *Panelist.*

USC Spatial Data Collection Using Drones (SSCI 220L)

2024 ***Charles Lehnen.*** Galápagos Tortoises. Los Angeles, CA, USA. *Guest lecturer.*

USC The Human Animal (HBIO200)

2022 ***Charles Lehnen.*** Galápagos Tortoises. Los Angeles, CA, USA. *Guest lecturer.*

Charles Darwin Foundation (CDF) Seminar Series

2018 ***Charles Lehnen****, *Paola Lahuatte, Bradley Sinclair, Charlotte Causton, George Heimpel.* Estudio Exploratorio de la Historia Natural de los Dípteros (moscas) de Galápagos. **.** Santa Cruz, Galapagos, Ecuador. *Oral presentation.*

Charles Darwin Foundation (CDF) Invertebrates Department Lunch Seminar

2018 ***Charles Lehnen****, Bradley Sinclair, Charlotte Causton, George Heimpel.* Ecología de los Dípteros de Galápagos: Estudio Exploratorio de la História Natural. Santa Cruz, Galapagos, Ecuador. *Oral presentation.*

St. Paul Optimist Club of Optimists International Monthly Presentation Series

2018 ***Charles Lehnen****, Bradley Sinclair, Charlotte Causton, George Heimpel.* Natural History of Galapagos Flies: An Exploratory Study. St. Paul, MN, USA. *Oral presentation.*

National Geographic & Lindblad Expeditions CDF Weekly Presentation Series (August)

2016 ***Charles Lehnen****, Bradley Sinclair, Charlotte Causton, George Heimpel.* Biological Control of *Philornis downsi*. Santa Cruz, Galapagos, Ecuador. *Oral presentation.*

National Geographic & Lindblad Expeditions CDF Weekly Presentation Series (July)

2016 ***Charles Lehnen****, Bradley Sinclair, Charlotte Causton, George Heimpel.* Biological Control of *Philornis downsi*. Santa Cruz, Galapagos, Ecuador. *Oral presentation.*

UMN College of Biological Sciences (CBS) Nature of Life BIOL 1806 Career Panel

2015 ***Charles Lehnen.*** Minneapolis, MN, USA. *Alumni panel.*

UMN CBS Student Board & the Biological Sciences Alumni Society BioBound 2014

2014 ***Charles Lehnen.*** Minneapolis, MN, USA. *Alumni panel.*

UMN CBS Alumni Relations and Development Common Time Career Panel

2014 ***Charles Lehnen.*** Minneapolis, MN, USA. *Alumni panel.*

UMN CBS Alumni Relations and Development Freshmen Career Panel

2014 ***Charles Lehnen.*** Minneapolis, MN, USA. *Alumni panel.*

UMN CBS Reflections from Abroad

2012 ***Charles Lehnen.**** Venezuela: Fall 2010. Minneapolis, MN, USA. *Poster.*

2026 ***Charles Lehnen****. Assessing Española Giant Tortoise introduction impact on Santa Fe Island with remote sensing and behavioral models.* ***North American Congress for Conservation and Restoration (NACCR)****. Oral presentation.* ***Student Presentation Award Finalist*** *https://xcdsystem.com/scbna/program/sNV3L7w/index.cfm?pgid=5961&print=1&printmode=1*

2026 ***Charles Lehnen****.* Activity Patterns of Introduced Giant Tortoises Across Time and Space on Santa Fe Island, Galápagos. **14th annual USC Graduate Student Government (GSG) Graduate Research Symposium**. *Oral lightning presentation*. ***2nd place winner.***

2025 ***Charles Lehnen****, Dr. Craig Stanford, Dr. Washington Tapia.* Machine Learning Classification of Camera Trap Imagery Reveals Selective Foraging on Plant Parts with Implications for Plant Reproduction and Landscape Heterogeneity. **2025 Annual Meeting of the Ecological Society of America.** *Oral presentation*. https://events.rdmobile.com/Lists/Details/2996841

2025 ***Charles Lehnen****, Dr. Craig Stanford, Dr. Washington Tapia.* Assessing Española Giant Tortoise introduction impact on Santa Fe Island with remote sensing and behavioral models. **Turtle Survival Alliance 23rd Annual Symposium.** *Oral presentation*. https://issuu.com/turtlesurvivalalliance/docs/23rd_annual_symposium_program

2024 ***Charles Lehnen****, Dr. Craig Stanford, Dr. Washington Tapia, Mariella Jorge.* Assessing Española Giant Tortoise introduction impact on Santa Fe Island with remote sensing and behavioral models. **2024 Annual Meeting of the Ecological Society of America.** *Oral presentation*. https://events.rdmobile.com/Lists/Details/2449201

2023 ***Charles Lehnen****, Andrea Morfin Valencia.* Ecology and Galapagos Tortoises. Giant Tortoises: The Newest Neighbors on Santa Fe Island. **Joint Educational Project (JEP) 2023 Seminar Series; *JEP and University of Southern California (USC)*.** *Oral presentation.*

2021 ***Charles Lehnen****, George Heimpel, Bradley Sinclair, Charlotte Causton.* Rearing sources of endemic and native Galapagos Diptera for non-target testing of *P. downsi* control agents. **International Workshop: Searching for solutions for the control of the Avian Vampire Fly, *Philornis downsi*, and the conservation of Galapagos landbirds; *CDF and GNP*.** *Oral presentation.*

2018 ***Charles Lehnen****, Paola Lahuatte, Bradley Sinclair, Charlotte Causton, George Heimpel.* Estudio Exploratorio de la Historia Natural de los Dípteros (moscas) de Galápagos. ***Agencia de Regulación y Control de la Bioseguridad y Cuarentena (ABG) para Galápagos.*** Santa Cruz, Galapagos, Ecuador. *Oral presentation.*

2018 ***Charles Lehnen****, Paola Lahuatte, Bradley Sinclair, Charlotte Causton, George Heimpel.* Estudio Exploratorio de la Historia Natural de los Dípteros (moscas) de Galápagos. ***Isabela Island Galapagos National Park (GNP; DPNG).*** Isabela, Galapagos, Ecuador. *Oral presentation.*

2013 ***Charles Lehnen.**** Beta-catenin stabilization to extend nTreg IL-2 dependent survival. **4th Annual Masonic Cancer Center Research Symposium; *UMN*.** Minneapolis, MN, USA. *Poster.*

Contributed Presentations

2025 *Mikaylie Wilson*,* ***Charles Lehnen****.* Ecological monitoring on Santa Fe Island, Galapagos. **Zoo and Wildlife Vet Nurses Australasia (ZWVNA).** Perth, Australia. *Oral presentation.*

2025 *Mikaylie Wilson*,* ***Charles Lehnen****.* Ecological monitoring on Santa Fe Island, Galapagos. **WReNNZ National Wildlife Conference.** Auckland, New Zealand. *Oral presentation.*

2025 *Alison Garcia*, Phil Kong,* ***Charles Lehnen****.* Introduced Ecological Proxies Impact Endemic Santa Fé Iguanas. 2025 USC Young Researchers Program Symposium. Los Angeles, CA *Poster presentation.*

2023 *Yan Yin Cheung*, Brian V. Brown, Vaughn Shirey, Teagan Baiotto, Austin Baker,* ***Charles Lehnen****, Jayme Lewthwaite, James Willoughby, Laura Melissa Guzman.* BioSCAN Survey Reveals Seasonal Trends of Insect Communities Across the Los Angeles Metropolitan Area. **Southern California Academy of Sciences 2023 Annual Meeting.** Newport Beach, CA. *Poster presentation.*

2020 *Charlotte E. Causton*, George E. Heimpel, Mariana Bulgarella, Ismael Ramirez, Rebecca Boulton, Gabriel Brito,* ***Charles Lehnen****, Denis Mosquera, Christian Sevilla.* ¿El Control Biológico, una Herramienta para el Manejo de la Mosca Parasítica Invasora Philornis downsi en las Islas Galápagos? Avances y Próximos Pasos. **2do Congreso de Control Biológico Aplicado; *Congreso de Control Biológico Aplicado*.** Bogotá, Colombia. *Invited keynote presentation.*

2020 *George E. Heimpel*, Charlotte E. Causton, Ismael E. Ramirez, Mariana Bulgarella, Rebecca A. Boulton,* ***Charles C. Lehnen****, Pamela Rueda-Cediel.* The prospect of using parasitoid releases against the invasive bird parasitic fly, *Philornis downsi* (Diptera: Muscidae) in the Galapagos Islands. **Entomology 2020: Virtual Annual Meeting; *Entomological Society of America*** Annapolis, MD, USA. *Invited presentation.*

Media

July 2026 — **“What a Great Teaching Idea: Supporting Student Engagement”** *USC Center for Excellence in Teaching (CET)*
https://cet.usc.edu/what-a-great-teaching-idea-ta-edition-supporting-student-engagement/?_thumbnail_id=10130

May 2026 — **“PhD candidate finds his field of dreams studying Galapagos tortoises”** *USC Dornsife College of Letters, Arts and Sciences News*
https://dornsife.usc.edu/news/stories/phd-student-studies-galapagos-tortoises-to-preserve-ecosystems/

May 2025 — **“The making of a field ecologist”** *University of Minnesota.*
https://twin-cities.umn.edu/news-events/making-field-ecologist

May 2025 — **“Dean's Scholars alumni spotlight: Charles Lehnen”** *University of Minnesota College of Biological Sciences.*
https://cbs.umn.edu/blog-posts/charles-lehnen

June 2024 — **“Reviving Ecosystems: How Galileo High Accuracy Service aids Galápagos tortoise reintroduction”** *European Union Agency for the Space Programme (EUSPA)*
https://www.euspa.europa.eu/newsroom-events/news/galileo-has-galapagos-tortoise

May 2024 — **“Tortoise Research: From USC to Galapagos”** *USC Dornsife*
https://www.youtube.com/watch?v=fTj72Qwsl0U

February 2024 — **“Do Giant Tortoises Make Good Neighbors?”** *EOS GNSS.*
https://eos-gnss.com/successes/galapagos

Winter 2022 — **"Where the Bats Go: New Research Uncovers Their Mysterious Journey"** *Esri ArcNews.*
https://www.esri.com/about/newsroom/arcnews/where-the-bats-go

Winter 2022 — **"Bat Conservation International: Successes"** *EOS GNSS.*
https://eos-gnss.com/successes/bat-conservation-international

Outreach and volunteership

March 2020 - August 2021 — **Spanish Interpreter.** *Alight (FKA American Refugee Committee)*
Supervisor: Annie Nolte-Henning

April - August 2015 — **Minnesota Herpetology Intern.** *In collaboration with Minnesota Herpetological Society; Minnesota Department of Natural Resources; Dakota County Natural Resources Management; Three Rivers Park District.*

November 2014 - August 2015	**Animal Care Crew (ACC) Member.** *Wildlife Rehabilitation Clinic (WRC) of Minnesota.* Supervisors: Karen Koeper, Bailey Stanard
October - December 2010	**Medical Volunteer.** *El Centro de Atención Médica Integral de la Universidad de Los Andes (CAMIULA) hospital* Supervisor: Dr. Rosario Gonzales, M.D.
September 2011 - January 2012	**Bilingual STEM tutor.** *El Colegio high school*
February - June 2010	**English as a Foreign Language Teacher.** *Family Opportunities for Living Collaboration (FOLC) adult education*
January - May 2009	**Reading Tutor.** *Carter G. Woodson Institute for Student Excellence (WISE) Charter School elementary school*

Mentorship

	Mentor. *USC Young Researchers Program (YRP) Mentor*
Summer 2025	Alison Garcia
Summer 2026	Arielle Escareno
	Field Director. *Santa Fé Island, Ecuador.*
Summer 2023	Simon Rouot; Juan Jose Gallardo; Mary-Cate Hyde, M.Sc.
Fall 2024	Tanja Laaser; Diego Rojas; Jeniffer Rodríguez;, Ilán Shalom
Spring 2025	(Jose) Felix Usiña, M.Sc; Mikaylie Wilson; Donna Samaniego; Felipe Quiñonez
	Field and laboratory supervisor. *Charles Darwin Foundation.*
2016 - 2019	Mery P. Lara, M.Sc; Hayden Wilke, M.Sc; Luna Heimpel
	Co-advisor. *UMN Undergraduate Research Opportunities Program (UROP)*
2014 - 2015	Riley Seitz, M.Sc; Lauren Li
	Mouse colony supervisor. *UMN Masonic Cancer Research Center*
2013 - 2015	Nara Paulsen P.A.

Selected GitHub Hosted Projects -

- African Megafauna Dataset from YouTube Livestreams: Designed a pipeline using computer vision (FasterRCNN_ResNet50_FPN) to capture, detect, crop, and manually classify African wildlife from livestream footage, laying groundwork for a large-scale classification dataset. *https://github.com/CharlesLehnen/YouTube_Trailcamera_Livestream_Classification_Tool*

- Enhancing the California Insect Barcoding Initiative (CIBI): Built an end-to-end pipeline to predict mantis distributions, integrating climatic rasters (NetCDF), ecoregions, human population layers,

GBIF biodiversity records, and county shapefiles; implemented linear regression, SVM, and decision-tree models via Colab/desktop notebooks, producing mapped predictions and CSV outputs to support barcoding of all California insect species.
https://github.com/CharlesLehnen/DSCI_550_Final_Project

- QGIS KML to DJI Pilot 2: Built a script/executable to convert QGIS-exported KML files into DJI Pilot 2–compatible formats, supporting cross-platform use (Windows/Linux/Python). https://github.com/CharlesLehnen/QGIS_KML_to_DJI_Pilot_2

- Garmin Instinct Topo Converter: Designed a graph-theory–based workflow that transforms DEMs into GPX topographic routes for Garmin Instinct watches, overcoming the device's lack of basemap support. This workflow generates navigable elevation-aware graphs usable directly on the watch. https://github.com/CharlesLehnen/Garmin_Instinct_Topo_Converter

- Lightweight-Charts for Freqtrade: Integrated TradingView's Lightweight-Charts into Freqtrade's Docker environment to enable interactive visualization of backtests and trading strategies, surpassing the difficult to use and unattractive built-in Plotly charts. https://github.com/CharlesLehnen/LightweightCharts-for-Freqtrade

- Google Drive Zip Folder Extractor: For when Google Takeout unavailable, built a Python/Windows executable tool to automatically extract and reconstruct folder hierarchies from split Google Drive zip downloads, including path sanitization and CSV-based renaming logs. https://github.com/CharlesLehnen/Google-Drive-Zip-Folder-Extractor

- Zotero Better BibTeX Output to Match Google Scholar: Created a Python tool that standardizes BibTeX keys exported from Zotero's Better BibTeX plugin to match Google Scholar citation formats so both tools can be used together. https://github.com/CharlesLehnen/Zotero_Better_BibTeX_Output_to_Match_Google_Scholar

Related Coursework -

Undergraduate Level : Advanced Biology & Lab, Medical Terminology, Human Physiology & Lab, Plant Physiology & Lab, Zoology & Lab, Animal Behavior & Lab, Tropical Ecology & Lab, Organic Chemistry & Lab, Biochemistry, Genetics, Cellular Biology, Bioinformatic Analysis, Multivariable Calculus, Applied Linear Algebra, Series and Sequences Theory

Graduate Level : Theory of probability & statistics; Principles of Programming for Data Science; Evolution & Human Biology; Spatial Data Acquisition; Principles of Spatial Data Analysis; Spatial Modeling with GIS; Center of Excellence in Teaching (CET) Future Faculty Teaching Institute certificate; Data Science at Scale; Statistical Methods for Environmental Scientists